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MIL-C-14381(QMC)
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MILITARY SPECIFICATION

CASES, FIRE DIRECTION SET, ARTILLERY

1. SCOPE

1.1 Scope. This specification covers fire direction set cases fabricated from olive drab cotton duck and webbing, felt padded to protect contents.

1.2 Classification. Fire direction set cases shall be of the following types as specified (see 6.2).

Type I—Case, Fire Direction Set No. 3, Artillery, 25,000 Meter Maximum Range.

Type II—Case, Fire Direction Set No. 4, Artillery, 15,000 Meter Maximum Range.

Type III—Case, Fire Direction Set No. 5, Artillery, 12,000 Meter Maximum Range.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

T-T-871 —Twine, Cotton, Wrapping.
V-T-276 —Thread, Cotton.
UU-P-268 —Paper, Kraft, Untreated Wrapping.
UU-T-111 —Tape; Paper, Gummed Sealing and Securing.
CCC-B-811 —Burlap; Jute.
CCC-C-429 —Cloth, Cotton, Osnaburg.
CCC-T-191 —Textile Test Methods.

MILITARY

MIL-A-140 —Adhesive, Water-Resistant, Waterproof Barrier Material.
MIL-T-2312 —Mildew Resistance and Moisture Resistance Treatment for Felt, Wool.
MIL-T-3530 —Treatment, Mildew-Resistant, for Thread and Twine.
MIL-B-13239 —Packaging and Packing for Overseas Shipment — Barrier Materials, Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-595 —Colors.

FSC 1290

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FED–STD–751 —Stitches, Seams, and Stitching.

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MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 —Marking for Shipment and Storage.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

A8636237

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

United States Army Standard Color Card for the Official Standardized Shades of Sewing Threads.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York 16, N. Y.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.1.1 *Thread, cotton, machine.* Thread shall have good colorfastness to chlorine bleaching, laundering and weathering in accordance with Specification V–T–276.

3.1.2 *Compound, lubricating, thread.* Lubricating compound shall be of the following proportions:

White mineral oil, light, sulphur free, specific gravity 0.830 to 0.860 .....One gallon

Copper–8–Quinolinolate solubilized type, 10 percent solution .....Four ounces

3.1.3 *Color.*

3.1.3.1 *Thread, cotton, machine.* Thread shall be vat dyed to match olive drab shade S–1, cable number 66022, of the United States Army Standard Color Card for the Official Standardized Shades of Sewing Threads (see 2.2).

3.1.3.2 *Cloth, cotton, duck.* Cloth shall be vat dyed olive drab to match color chip No. 34102, Table IV, of Standard FED–STD–595.

3.1.4 *Treatment, mildew and moisture resistant.*

3.1.4.1 *Thread, cotton, machine.* Thread shall be treated in accordance with Type 1, Class 1 of Specification MIL–T–3530.

3.1.4.2 *Cloth, felt.* Felt shall be treated in accordance with Type III of Specification MIL–T–2312.

3.2 **Fabrication.** The cases shall be manufactured in accordance with Drawing A8636237, and drawings pertaining thereto. Requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.2.1 *Stitching, machine.*

3.2.1.1 *Type of stitching.* Stitching shall conform to Specification FED–STD–751 as follows:

All stitching except overedge stitching

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shall be Type 301, 7 to 8 stitches to the inch.

Overedge stitching shall be Type 304, 14 to 16 stitches to the inch.

3.2.1.1.1 *Type 301 stitching.* Thread breaks in stitching shall be back-stitched not less than 1 inch at each break. Stitching shall be backstitched at the ends to prevent raveling, except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained properly so there will be no loose stitching and that the lock will be embedded in the center of the materials sewed.

3.2.1.1.2 *Type 304 stitching.* Thread breaks in stitching shall be overstitched not less than ½ inch at each break. Thread tension shall be maintained properly so that there will be no loose stitching.

3.2.2 *Thread type.* Thread Type IB3, size 12/4 shall be used for needle or top thread and Type IA3, size 12/4 for bobbin in machine on all stitching except overedge stitching. Thread Type IB3, size 24/4 shall be used for needle and Type IA3, size 24/4 for bobbin in machine on overedge stitching (see 3.1.1).

3.2.3 *Lubrication of thread.* Use of lubricating compound specified in 3.1.2 will be permitted, if necessary, to reduce thread breakage in production sewing.

3.2.3.1 *Application of lubricating compound.* The thread may be lubricated during the sewing operation by running it against a felt pad saturated with the lubricating compound, or the full package of thread may be immersed in the same compound before sewing.

3.2.4 *Setting of ball type end clips.* The webbing shall be inserted the full depth of the clips. The clips shall be securely attached without cutting the webbing and the webbing shall be flat at the point of entrance.

3.3 Marking.

3.3.1 The letters "US", nomenclature of the item, and identification marks shall be printed or stenciled in the size characters and locations shown on the drawings. Identification marks shall be as follows:

Approved nomenclature of item.
Stock number of case.
Number and date of contract.
Name of contractor.

3.3.2 Black printing ink or stencil ink, light and water spray fastness, shall be used for all marking.

3.4 Workmanship. The finished case shall be clean, free from any defect which may affect appearance or serviceability, and shall be representative of quality workmanship and application of appropriate manufacturing processes in fulfillment of requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Except where otherwise specified herein, the contractor's inspection system shall be in accordance with Specification MIL–I–45208.

4.3 Lot formation. When inspection lot sizes and lot formations are applicable, they

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shall be in accordance with Standard MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.3.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

4.4 **Inspection provisions.** Components and sub-assemblies shall be examined in accordance with the inspection provisions contained in Specification MIL–F–13926 and 4.4.1 herein. Test and control of materials not covered by Supplementary Quality Assurance Provisions (SQAP's) or Section 4 of Specification MIL–F–13926 shall be in accordance with "Test Data" provisions of Specification MIL–I–45208. Inspection related to Section 3 of this specification shall be performed in accordance with 4.4.2. The tabulated Classification of Defects listed herein shall constitute the minimum inspection to be performed by the contractor prior to Government acceptance. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.4.1 *Inspection of components.* All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of the carrying cases shall be inspected in accordance with the written inspection system required by Specification MIL–I–45208. These characteristics are generally defined in Specification MIL–F–13926 as Class 1 or Class 2 Defects.

4.4.1.1 *Component sampling inspection.* Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with the sampling plan tables specified in Specification MIL–F–13926.

4.4.2 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of Section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with Standard MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in Table I may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

4.4.2.1 *Disposition of rejected lots.* Rejected lots shall be screened in conformance with 4.3.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.4.2 and Standard MIL–STD–105.

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TABLE I. *Classification of defects.*

Use Inspection Level II in Table III with Sampling Plan Table IV-A of Standard MIL-STD-105.

Critical: None defined.

	Req't par.	QA par.	Test method
Major: AQL 1.0 percent defective.			
101. Fabrication non-conforming	3.2	4.6.2	Std meas equip (SME)-Visual
102. Improper stitching	3.2.1	4.6.2.1	SME-Visual
103. Improper type thread	3.2.2	4.6.2.1	SME-Visual
104. Setting of end clips non-conforming	3.2.4	4.6.2.3	Visual—Tactile
105. Workmanship	3.4	4.6.4	Visual—Tactile
Minor: AQL 4.0 percent defective.			
201. Improper application of lubrication thread.	3.2.3	4.6.2.2	Visual

Note. The above tests shall be conducted at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit) in accordance with the quality assurance paragraph (QA par.) listed.

4.4.3 *Control inspection.* The first 2 carrying cases produced, and 3 selected at random from every 100 subsequently produced (or from each month's production, whichever occurs first) shall represent the control samples for control inspection. Control samples shall be subjected to the examinations and tests specified herein to determine compliance with 3.1 and 3.3. The examinations and tests shall be conducted by the contractor under surveillance of the Government representative at the contractor's facility, or Government approved laboratory, using test equipment authorized for use by the contract, or otherwise approved by the contracting officer.

4.4.3.1 *Control sample failure.* Should a control sample fail to meet the requirements specified in 4.4.3, the contractor shall cease those operations causing rejection until necessary corrections have been made by the contractor and approved by the Government. When production has resumed, inspection of those requirements that have previously caused rejection shall be performed on a 100 percent basis until 5 consecutively produced carrying cases meet these requirements. When this is accomplished, control inspection (4.4.3) may be continued.

4.5 Inspection equipment. Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.5.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of permissible tolerances except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6 Test methods and procedures.

4.6.1 *Materials.* The materials shall be inspected in accordance with 4.4.3 and the quality assurance provisions of the applicable specifications to determine compliance with 3.1 and 3.2.1. The Government shall reserve the right to require materials or a chemical analysis of materials or certifica-

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tion thereof to insure quality of material in accordance with the applicable specifications.

4.6.1.1 *Lubricating compound.* Acceptance of compound to be used may be by certification to insure conformance with 3.1.2. Data obtained from laboratory analysis as well as evidence of correct proportions used shall be available at all times.

4.6.1.2 *Color.* The carrying case shall be visually inspected for correctness of color to determine compliance with the applicable specifications specified in 3.1.3.

4.6.1.3 *Treatment, mildew and moisture resistant.* The test specimens, or carrying cases, shall be inspected in accordance with the quality assurance provisions of the applicable specification to determine compliance with 3.1.4.

4.6.2 *Fabrication.* The carrying case shall be subjected to a visual, tactile and dimensional inspection to determine conformance to 3.2 and the applicable drawings.

4.6.2.1 *Stitching and thread.* The stitching and thread applied to the case shall be inspected in conformance with the procedures specified in Federal Standard FED–STD–751 for stitching in conformance with 3.2.1, and that the respective type and size thread specified meets the requirements of 3.2.2.

4.6.2.2 *Lubrication of thread.* When lubrication is used, it shall be determined from data available from certification that the lubrication material meets the requirements of 3.1.2, and shall be visually inspected to insure compliance with 3.2.3.1.

4.6.2.3 *Setting of ball type end clips.* Clips shall be subjected to a visual and tactile examination for conformance with 3.2.4. Clips shall be securely attached and measured for the specified distance. A manual pull with a minimum force of 20 pounds shall be applied to the end clip and webbing by gripping the webbing with one hand and the clip end with the fingers of the other hand or other equivalent means to obtain the required force. Any indication of looseness or movement of the clip on the webbing shall be cause for rejection.

4.6.3 *Marking.* Marking and identification shall be visually inspected for legibility and accuracy. Size and position of the markings shall be measured with standard measuring equipment. Identification and markings shall meet the requirements specified in 3.3.1 acceptable upon certification of the contractor that the requirements of 3.3.2 have been met.

4.6.4 *Workmanship.* The case shall be given an overall visual and tactile examination for conformance with the requirements specified in 3.4.

4.7 Packaging, packing and marking. Quality assurance provisions for packaging, packing and marking shall be in accordance with Section 5, utilizing the test methods and procedures specified in Specification MIL–P–14232.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Each carrying case shall have two ounces of napthalene flakes uniformly distributed over the entire area of the felt before folding.

5.1.1.1 *Type I and II.* Each case shall be laidout flat, with shoulder straps, shoulder pads, and billets tucked inside, and folded in increments of one third the length. Each folded case shall be tied girthwise with suitable cotton cord or twine, approximately 6 inches in from each end. Six cases, of one type only, shall then be neatly and evenly stacked one on top of another and securely

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tied in three places with suitable cotton cord or twine. The tying shall be applied girthwise 6 inches in from each end with the third tie centered.

5.1.1.2 *Type III.* Each case shall be laid out flat and folded once lengthwise so that the top of the handles are even with the bottom edge of the case. Six folded cases shall be processed in the same manner as that specified for the Type I and Type II cases.

5.1.1.3 *Kraft paper wrap.* Each bundle (6 cases), tied as specified, shall be wrapped in 50-pound Kraft paper conforming to Grade B of Specification UU–P–268. The Kraft paper wrap shall be held in place with not less than 3-inch gummed Kraft paper tape conforming to Class 2 of Specification UU–T–111.

5.1.2 *Level B.* Cases shall be packaged as specified in 5.1.1.

5.1.3 *Level C.* Cases shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Each bundle wrapped in Kraft paper as specified in 5.1.3 shall be overwrapped in waterproof barrier material conforming to Type B–2 of Specification MIL–B–13239. All seams and closure shall overlap not less than ¾ inch and shall be sealed with an adhesive conforming to Type II, Grade B, Class 2 of Specification MIL–A–140.

5.2.1.1 Each bundle shall then be firmly and compactly wrapped in cotton osnaburg cloth weighing not less than 6.8 ounces per square yard, conforming to Specification CCC–C–429, or in burlap weighing 10 ounces per linear yard of 40-inch width material conforming to Specification CCC–B–811. The wrapping shall be sewn with 16 ply, type II, Class I, cotton twine, conforming to Specification T–T–871, with approximately one stitch to the inch and every third stitch knotted. Five inches of surplus covering shall be gathered together on each of the four corners and securely sewn into ears for handles. Alternatively the osnaburg cloth or burlap may be secured in tubbing form, wire tied both ends or pre-sewn one end and wire tied the open end.

5.2.2 *Level B.* Cases shall be packed as specified in 5.2.1, except that the waterproof barrier material overwrap shall not be required.

5.2.3 *Level C.* The cases shall be packaged and packed in a manner acceptable to common carriers for safe transportation to point of destination specified in shipping instructions, at the lowest transportation rate for such supplies.

5.3 Marking. The bales shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The cases covered by this specification are designed to accommodate and protect fire direction equipment used by field artillery.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Type of case required (see 1.2).

(c) Applicable stock number.

(d) Selection of applicable levels of preservation, packaging and packing.

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